

UMA071: NUMERICAL OPTIMIZATION

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Course Objectives: The objective of this course is to provide students with a comprehensive understanding of optimization techniques, both linear and nonlinear, and their applications in various fields. Students will gain knowledge of fundamental concepts, methods, and algorithms in optimization and develop the skills necessary to formulate and solve optimization problems. Through this course, students will learn to apply optimization models and techniques to realworld problems, enhancing their analytical and problem-solving abilities.

Unit 1: Introduction: Definition of optimization, types of optimization models (linear vs nonlinear, constraint vs unconstraint optimization models), geometry of linear programming

Unit 2: Linear optimization: Basic Feasible Solutions and Algebraic Method, Simplex method, its variants, and special cases, duality for Linear Programming Problems (LPP), branch and bound method for Integer Linear Programming Problems (ILPP)

Unit 3: Unconstrained Optimization: Definiteness of matrices, optimality conditions for unconstrained optimization, special case of quadratic functions, convex functions and their properties, applications of quadratic functions to least square approximation

Unit 4: Search Techniques: Steepest descent method, conjugate gradient method, Newton and Quasi-Newton methods, Fletcher and Reeve's method, David-Fletcher-Powell method

Unit 5: Constrained Optimization, Convex optimization, KKT conditions for general inequality and equality constrained problems, Frank and Wolfe's method, Rosen gradient projection method, Penalty and barrier functions, Convex quadratic problem and its application to classification by linear separators, convex optimization in support vector machines(SVM).

Course Learning Outcomes (CLOs) / Course Objectives (COs):

Upon Completion of this course, the students would be able to:

1. Formulate classify optimization problems.
2. Apply the Simplex Method and its variants for solving the Linear programming problems, and analyse the Integer Programming Problems.
3. Implement Search Techniques in Unconstrained Optimization.
4. Evaluate Constrained Optimization Problems.
5. Demonstrate Proficiency in Using Software Tools such as MATLAB or other relevant